GRADE 10 · TERM 2

Decisions, Text & Networks

A pencil-and-paper workbook: think it through first, learn the Delphi, then prove you can do it.

Name: _____

Class: _____ Date: _____

How to use this workbook

1. Work with a **pencil**. Mistakes are how we learn: rub out and try again.
2. Start each unit with **Think First**. Try it on your own before you read the explanation.
3. Read **Learn It**, then do the activities. Trace tables and code are written **by hand**.
4. At the end of each unit, tick the "I can" boxes you are sure of. Write down anything you still want to ask.
5. The **checkpoint test** is done under test conditions, with no notes.

In this workbook

1. Everyday decisions
2. IF…THEN…ELSE and nested IF
3. AND, OR, NOT and IN
4. CASE and MessageDlg
5. Strings and validation
6. Devices, ports and networks
7. Term checkpoint test

> **Matches the Annual Teaching Plan:** Term 2 — full IF…THEN…ELSE and nested IF; AND, OR, NOT and IN; CASE and MessageDlg; Length, UpperCase, UpCase and Trim; input validation; FormCreate and Activate; input, output and storage devices; the system unit; ports; PAN, HAN, LAN and WAN; client-server and peer-to-peer networks.

UNIT 1 Everyday decisions

You make decisions all day: Is it raining? Do I have money for a pie? A program does the same, but it can only ask questions with a YES or NO answer. In this unit you will practise decisions on paper before you meet them in Delphi.

By the end of this unit I can: (tick each box when you are sure)

- ☐ turn a rule into a question with a Yes/No answer
- ☐ follow and draw a flowchart that has a decision in it
- ☐ explain what a nested decision is
- ☐ complete truth tables for AND, OR and NOT
- ☐ decide if a combined condition is True or False

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

1.1 Who gets the free sweet? [4]

Mama Thandi runs a spaza shop. She has two different offers and she wants to know who qualifies for a free sweet.

Offer A: you must buy bread **AND** milk. **Offer B:** you must buy bread **OR** milk (or both). Write Yes or No in each box.

The customer buys…	Offer A (AND)	Offer B (OR)
Bread and milk		
Bread only		
Milk only		
Nothing		

In your own words: when is **AND** stricter than **OR**?

LEARN IT

Conditions and flowcharts

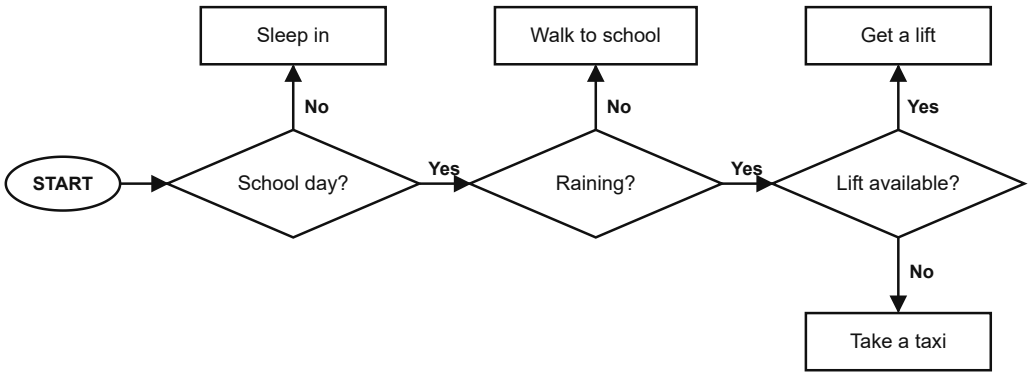
A **condition** is a question that can only be answered **True** or **False** (Yes or No): "Is the mark 50 or more?" is a condition; "What is the mark?" is not. A **flowchart** draws the steps of an algorithm with arrows:

- **Oval**: where you START or STOP. **Rectangle**: an action (input, calculate, display).
- **Diamond**: a decision, with one arrow in and **two** arrows out, labelled Yes and No.

A decision placed **inside** the Yes or No path of another decision is **nested**. Nesting lets you ask a second question only when the first answer needs it.

1.2 Follow the flowchart [4]

Lerato decides how to get to school using this flowchart. Follow the arrows and write what she does in each situation.



Situation	What does Lerato do?
Saturday, raining	
Monday, sunny	
Tuesday, raining, nobody can give her a lift	
Wednesday, raining, her aunt can give her a lift	

Why does Lerato never ask "Is it raining?" on a Saturday?

1.3 Draw a flowchart [6]

Draw a flowchart on the grid for this algorithm. Use the correct shapes and label every arrow that leaves a diamond.

Input a mark (iMark). If the mark is 50 or more, then check again: if it is 80 or more display 'Distinction', otherwise display 'Pass'. If the mark is below 50 display 'Not yet'. Then stop.

LEARN IT

AND, OR and NOT

Programmers join conditions with three words:

- **A AND B** is True only when **both** A and B are True.
- **A OR B** is True when **at least one** of them is True.
- **NOT A flips** the answer: NOT True is False, NOT False is True.

A **truth table** lists every possible combination of True and False and shows the result. With two conditions there are 2 × 2 = 4 rows.

1.4 Truth table: AND, OR and NOT [8]

Complete the table (T = True, F = False). Work out NOT A first; then use it for the last column.

A	B	A AND B	A OR B	NOT A	(NOT A) AND B
T	T				
T	F				
F	T				
F	F				

1.5 True or False? [6]

Thabo is **16** years old (iAge = 16), has **R18** (rMoney = 18) and it **is** break (bBreak = True). Circle True or False for each condition.

1	(iAge >= 14) AND (rMoney >= 20)	**True**	**False**
2	(iAge >= 14) OR (rMoney >= 20)	**True**	**False**
3	NOT bBreak	**True**	**False**
4	(rMoney > 10) AND (NOT bBreak)	**True**	**False**
5	(iAge = 15) OR bBreak	**True**	**False**
6	NOT ((iAge < 18) AND (rMoney > 100))	**True**	**False**

UNIT 2 IF…THEN…ELSE and nested IF

In Term 1 you used a simple IF. Now you will give the program two paths (IF…ELSE), put decisions inside decisions, and learn why the *order* of your tests matters.

By the end of this unit I can: (tick each box when you are sure)

- ☐ write IF…THEN…ELSE correctly, with no semicolon before ELSE
- ☐ use BEGIN…END when a branch has more than one statement
- ☐ write and trace a nested IF
- ☐ put the tests of an IF chain in the correct order
- ☐ find the usual IF mistakes in code

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

2.1 Mr Khumalo's sweets

Mr Khumalo hands out sweets for marks. His rules are on the board, in this order: **(1)** 30 or more: 1 sweet. **(2)** 50 or more: 2 sweets. **(3)** 80 or more: 3 sweets. He reads them from the top and **stops at the first rule that is true**. Thandi scored 85. How many sweets does she get? Is that fair? Rewrite the three rules in an order that works for every learner.

LEARN IT

IF…THEN…ELSE

An IF statement chooses between **two** paths. If the condition is True the statement after **then** runs; if it is False the statement after **else** runs. Never both. **Two traps:** (1) There is **no semicolon** before **else**: a semicolon ends the whole IF, so the else has nothing to belong to and Delphi reports an error. (2) A branch with **more than one statement** must be wrapped in **begin … end**; otherwise only the first statement belongs to the IF.

Delphi

```
if iMark >= 50 then
begin                                  // two statements: begin ... end
  lblResult.Caption := 'Pass';
  btnCertificate.Enabled := True;
end                                    // NO semicolon before else
else
  lblResult.Caption := 'Not yet';      // one statement: no begin needed
```

LEARN IT

Nested IF and the ELSE IF chain

Put an IF inside the **else** (or the **then**) of another IF to ask a second question. Delphi tests from the top and takes the **first True**; after that it skips the rest. So the **order matters**: test the most demanding condition first.

Delphi

```
// a chain: the order of the tests matters
if iMark >= 80 then
  sResult := 'Distinction'
else
  if iMark >= 50 then
    sResult := 'Pass'
  else
    sResult := 'Not yet';

// a second question is only asked when the first is True
if bRegistered then
begin
  if rFine = 0 then
    sMsg := 'You may borrow'
  else
    sMsg := 'Pay your fine first';
end
else
  sMsg := 'Please register';
```

2.2 Be the computer: pie prices [8]

A pie costs less when you buy more. Complete the table for each value of iPies. Write *Yes* or *No* for each test, or a dash (-) if the test is never reached.

Delphi

```
if iPies >= 10 then
  rPrice := 11
else
  if iPies >= 5 then
    rPrice := 12
  else
    rPrice := 14;
rTotal := iPies * rPrice;
```

iPies	iPies >= 10 ?	iPies >= 5 ?	rPrice	rTotal
3				
5				
10				
12				

2.3 Debug: two broken IFs [8]

A does not compile: find **two** errors. **B** compiles, but a learner who scores **50** should pass and **only** a passing learner may see the certificate button: find **two** errors. Write the corrected code.

```delphi
// A
if iAge >= 18
  lblStatus.Caption := 'Adult';
else
  lblStatus.Caption := 'Learner';

// B
if rMark > 50 then
  lblResult.Caption := 'Pass'
else
  lblResult.Caption := 'Not yet';
btnCertificate.Visible := True;
```

2.4 Write it: taxi fare [6]

A taxi charges **R10** for a child under 12, **R12** for a pensioner (60 or older) and **R20** for everyone else. The age is in the spin edit **sedAge**. Write the code to work out iFare and show it in **lblFare** as, for example, *R20*.

2.5 Debug: wrong order [4]

The symbol should be A for 70 or more, B for 50 to 69, C for 30 to 49 and D below 30. The code is meant to do that. What symbol does it give for marks of **75**, **55** and **20**? Then rewrite the tests so that it works.

Delphi

```
if iMark >= 30 then
  sSymbol := 'C'
else
  if iMark >= 50 then
    sSymbol := 'B'
  else
    if iMark >= 70 then
      sSymbol := 'A'
    else
      sSymbol := 'D';
```

2.6 Write it: library books [6]

A learner may borrow a book only if registered (checkbox **chkRegistered**). If registered, check the fine typed in **edtFine** (a real number): a fine of 0 shows *You may borrow a book* in **lblMsg**, otherwise *Please pay your fine first*. A learner who is not registered sees *Please register first*. Use a nested IF.

UNIT 3 AND, OR, NOT and IN

Real rules usually have more than one part: "older than 14 *and* has permission". In this unit you will combine conditions in Delphi, use brackets so they mean what you intend, and meet the IN operator.

By the end of this unit I can: (tick each box when you are sure)

- ☐ combine conditions with AND, OR and NOT
- ☐ explain why each condition needs its own brackets
- ☐ work out the order in which NOT, AND and OR are applied
- ☐ use IN with a set or a range
- ☐ write a condition from a sentence

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

3.1 The tuck shop sign

A hand-written sign at the tuck shop reads: **"Pie AND chips OR cooldrink: R20."**
Nomsa has only a cooldrink. Zinzi has a pie and chips. Hendrik has a pie and a cooldrink. Who might the shopkeeper let through, and why do Nomsa and Hendrik argue? Show, with brackets, the **two** different meanings of the sign.

LEARN IT

Brackets, and the order of AND, OR and NOT

Delphi applies **NOT first**, then **AND**, and **OR last**. It does this even if you wrote the OR first, just like multiplication happens before addition in maths. Comparisons such as *iAge >= 14* are applied *after* AND/OR, so **every comparison needs its own brackets**; otherwise Delphi tries to AND the numbers together and gives an error. Always use brackets: they make the meaning clear to you and to the computer.

Delphi

```
// WRONG, does not compile:  if iAge >= 14 and iAge <= 19 then
if (iAge >= 14) and (iAge <= 19) then
  ShowMessage('Teenager');

if (bPaid) or (bSRC) then
  ShowMessage('Entry allowed');

if not (iMark >= 50) then      // same as iMark < 50
  ShowMessage('Not yet');

bResult := bA or bB and bC;    // means bA or (bB and bC)
bResult := (bA or bB) and bC;  // a different question!
```

LEARN IT

The IN operator

IN checks whether a value is in a **set**, written in square brackets. It replaces long chains of OR. A set can list values, give a range with **..**, or mix both. It works with whole numbers and characters (not with real numbers or strings).

Delphi

```
if cLetter in ['A', 'E', 'I', 'O', 'U'] then
  ShowMessage('Vowel');

if iMonth in [6..8] then          // June to August
  ShowMessage('Winter');

if not (iSeat in [1..15, 40..55]) then
  ShowMessage('Seat must be 1-15 or 40-55');
```

3.2 Truth table: school trip [8]

A learner may go on the trip if they have paid **and** have a permission slip **and** are **not** on detention: *(bPaid AND bPermission) AND (NOT bDetention)*. Complete the table (T = True, F = False).

bPaid	bPermission	bDetention	NOT bDetention	May go?
T	T	T		
T	T	F		
T	F	T		
T	F	F		
F	T	T		
F	T	F		
F	F	T		
F	F	F		

3.3 Predict: the precedence trap [3]

Work out the value of each Boolean. Then say in one sentence why they are different.

```Delphi
bA := True;
bB := False;
bC := False;
bResult1 := bA or bB and bC;
bResult2 := (bA or bB) and bC;
```

3.4 Evaluate with these values [6]

Use **iMark = 62**, **cGrade = 'C'**, **bPaid = True** and **iDay = 6**. Circle True or False for each condition.

1	(iMark >= 50) AND bPaid	**True**	**False**
2	(iMark IN [40..59]) OR (cGrade = 'B')	**True**	**False**
3	cGrade IN ['A'..'C']	**True**	**False**
4	NOT bPaid	**True**	**False**
5	NOT (iDay IN [1..5])	**True**	**False**
6	(iMark < 50) OR NOT bPaid	**True**	**False**

3.5 Debug: three broken conditions [6]

Each condition has a mistake. Say what is wrong and write it correctly.

```Delphi
1.  if iAge >= 14 and iAge <= 19 then
2.  if cLetter = 'A' or 'E' or 'I' then
3.  if not iMark >= 50 then
```

3.6 Write the conditions [8]

Write a Delphi condition (the part between *if* and *then*) for each sentence.
(a) The age **iAge** is from 14 to 16, including both.
(b) Entry is free if **iAge** is under 6 **or** 60 and over.
(c) The grade **iGrade** is **not** 8, 9 or 10.
(d) The character **cKey** is a digit from 0 to 9.

3.7 Write it: matric dance ticket [7]

A ticket costs R150. Only Grade 11 and 12 learners may buy one (the grade is in the spin edit **sedGrade**). A learner gets **R30 off** if they paid early (**chkEarly** is checked) **or** are on the SRC (**chkSRC** is checked). Show the price in **lblPrice** as *R120*. A learner in any other grade sees *Not eligible* in the label. Use an integer variable **iPrice**.

UNIT 4 CASE and MessageDlg

When one value can lead to many different outcomes, a long IF chain gets untidy. CASE is the neat alternative. You will also learn to ask the user a Yes/No question with MessageDlg.

By the end of this unit I can: (tick each box when you are sure)

- ☐ decide whether IF or CASE is the better tool
- ☐ write a CASE with single values, lists, ranges and an else part
- ☐ trace a CASE statement
- ☐ find common CASE mistakes
- ☐ ask a question with MessageDlg and react to the answer

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

4.1 The taxi rank fare board [4]

The board at the taxi rank says: **Zone 1: R12 | Zones 2 and 3: R18 | Zones 4 to 6: R25.** Write the fare for each passenger.

A passenger travels to zone...	Fare
2	_____
6	_____
7	_____

What should the driver do about zone 7?

_____

LEARN IT

The CASE statement

CASE looks at **one** value (the *selector*) and jumps to the line that matches. A line holds one value, several values separated by commas, or a range written with two dots. The optional **else** catches everything else.

- The selector must be a **whole number or a character**: not a real number, not a string.
- Ranges must not overlap: a value may not appear in **two** lines.
- CASE has **no begin**, but it finishes with **end**.

Delphi

```
case iZone of
  1    : iFare := 12;
  2, 3 : iFare := 18;
  4..6 : iFare := 25;
else
  iFare := 40;
end;
```

4.2 IF or CASE? [6]

Circle the better tool for each job.

1	Turn a mark from 0 to 100 into a symbol from 1 to 7.	**IF** **CASE**
2	Check that the learner is 14 or older **and** has paid.	**IF** **CASE**
3	Turn the day number 1 to 7 into a day name.	**IF** **CASE**
4	Decide whether a real-number price is above R100.	**IF** **CASE**
5	Choose the taxi fare from the zone number.	**IF** **CASE**
6	Check whether the text in edtName is the word Admin.	**IF** **CASE**

4.3 Fill in the blanks [6]

The selector of a CASE must be an _____ value such as an Integer or a Char. A range of 4 to 6 is written 4 ____ 6. The part that runs when nothing matches is called _____. A CASE statement ends with the word _____ but it has no _____. A value may not appear in _____ different lines.

4.4 Be the computer: taxi zones [8]

Use the CASE statement in *Learn It*. For each zone, write which line matches (*1*, *2, 3*, *4..6* or *else*) and the value of iFare.

Delphi

```
case iZone of
  1    : iFare := 12;
  2, 3 : iFare := 18;
  4..6 : iFare := 25;
else
  iFare := 40;
end;
```

iZone	Line that matches	iFare
1		
3		
4		
7		

4.5 Predict: sorting letters [3]

What is the value of sGroup after the code runs when cLetter is **'M'**? What if it is **'Z'**? What if it is the lowercase **'c'**? (Capital letters come before lowercase letters in the alphabet of the computer.)

Delphi

```
case cLetter of
  'A'..'F' : sGroup := 'Group 1';
  'G'..'L' : sGroup := 'Group 2';
  'M'..'R' : sGroup := 'Group 3';
else
  sGroup := 'Group 4';
end;
```

4.6 Debug: the day types [4]

This code should say whether iDay (1 = Monday … 7 = Sunday) is a school day, a weekend day or invalid. It does not compile. Find **two** errors.

Delphi

```
case iDay of
  1..5 : sType := 'School day';
  5..7 : sType := 'Weekend';
else
  sType := 'Invalid day';
```

LEARN IT

MessageDlg: ask the user a question

ShowMessage only tells the user something. **MessageDlg** *asks* and gives you the answer. It is a function with four parts: the **text**, the **icon** (mtConfirmation, mtInformation, mtWarning, mtError), the **buttons** (mbYesNo, mbOKCancel, mbYesNoCancel) and a **0** for help (always 0 for now). It returns which button was clicked: **mrYes**, **mrNo**, **mrOK** or **mrCancel**. Compare the answer in an IF.

Delphi

```
if MessageDlg('Delete all the entries?', mtConfirmation, mbYesNo, 0) = mrYes then
  memOut.Clear
else
  ShowMessage('Nothing was deleted');
```

4.7 Write it: mark to symbol **[6]**

The mark is in **iMark** (0 to 100). Use a CASE statement to put the symbol in **sSymbol**: 80–100 is *7*, 70–79 is *6*, 60–69 is *5*, 50–59 is *4*, 40–49 is *3*, 30–39 is *2* and below 30 is *1*.

4.8 Write it: leaving the program **[6]**

When the user clicks **btnExit**, ask *Do you really want to quit?* with a question icon and Yes/No buttons. If the answer is Yes, close the form with **Close**. Otherwise show *Welcome back!* with ShowMessage.

UNIT 5 Strings and validation

Programs must cope with users who type spaces, capitals and nonsense. Here you will measure and tidy text, check data *before* you use it, and decide when set-up code should run.

By the end of this unit I can: (tick each box when you are sure)

- ☐ use Length, UpperCase, UpCase and Trim and say what each returns
- ☐ explain why a program must validate input
- ☐ write checks for empty text, a range and a negative number before Sqrt
- ☐ explain the difference between FormCreate and Activate
- ☐ write a short event handler

One thing I want to ask my teacher: _____

5.1 Count, trim, capitalise [3]

A learner typed a name with a space before and after it. The word below is stored in **sName** and every box is one character (the empty boxes are spaces). (a) How many characters are there in all (**Length**)? (b) What would **Trim** leave, and what is its length? (c) Write the result of **UpperCase** applied to the trimmed name.

1	2	3	4	5	6	7	8	9	10
	A	y	a		D	u	b	e	

THINK FIRST · no computer needed

5.2 The form checker [3]

A school clerk receives this subject-choice form. Which of these entries should she reject, and which rule would she use for each (empty, out of range, wrong length)?

Field	What the learner wrote
Name	(left blank)
Grade	17
Cell number (10 digits)	082 55

LEARN IT

Four string tools

Length(s) counts the characters (spaces included) and returns a whole number. **UpperCase(s)** turns a whole *string* into capitals. **UpCase(c)** does the same for *one character*. **Trim(s)** removes spaces at the start and end of a string. Each one **returns a new value**: the original does not change unless you store the result back in a variable. Position numbers start at 1, so s[1] is the first character.

Delphi

```
sName := Trim(edtName.Text);        // '  aya ' becomes 'aya'
iLen := Length(sName);              // 3
sBig := UpperCase(sName);           // 'AYA'
cFirst := UpCase(sName[1]);         // 'A'  (one Char, so UpCase)
sName := UpperCase(Trim(edtName.Text));   // functions can be nested
```

5.3 What do the functions give? [5]

sName holds **' Zanele Mkhize '**, which starts with two spaces and ends with one. Write the result of each expression. Put text answers in single quotes.

Expression	Result
Length(sName)	
Trim(sName)	
Length(Trim(sName))	
UpperCase(Trim(sName))	
UpCase(sName[4])	

LEARN IT

Validate before you process

Garbage in, garbage out. A program must **check** data *before* it uses it. Common checks: **presence** (text is not empty), **range** (a number is within limits) and **length** (the right number of characters). Use IF, and use **Exit** to leave the event handler when the data is bad so that nothing else runs. A classic: Sqrt crashes on a negative number, so test the number first.

Delphi

```
rNum := StrToFloat(edtNum.Text);
if rNum < 0 then
  ShowMessage('You cannot take the square root of a negative number')
else
  lblRoot.Caption := FloatToStrF(Sqrt(rNum), ffFixed, 8, 2);

if Trim(edtName.Text) = '' then
begin
  ShowMessage('Please type your name');
  Exit;                        // stop here, do nothing else
end;
```

5.4 Be the computer: ticket validation [8]

A ticket costs R45. The rule says between 1 and 6 tickets may be ordered. Complete the table for each value typed.

Delphi

```
iQty := StrToInt(edtQty.Text);
if (iQty < 1) or (iQty > 6) then
  sMsg := 'Invalid'
else
  sMsg := 'Total R' + IntToStr(iQty * 45);
```

iQty	iQty < 1	iQty > 6	sMsg
0			
6			
7			
3			

5.5 Debug: the greeting [6]

This code should greet the learner by name and show the first letter in capitals. Find **three** errors and rewrite the code correctly.

Delphi

```
sName := Trim(edtName.Text);
if sName = 0 then
  ShowMessage('Please type your name');
cInitial := UpperCase(sName[1]);
lblHi.Caption := 'Hello ' + sName + ', your initial is ' + cInitial;
```

5.6 Write it: when the number is negative [5]

The learner types a number in **edtNum** and clicks the button to get the square root in **lblRoot**. Write the code so the program does **not** crash when the number is negative: show the message *Negative numbers have no square root* instead.

LEARN IT

FormCreate and Activate: when does code run?

Windows creates your form once and then you can click away and back. Two **events** let you run code at the right moment. **FormCreate** happens **once**, when the form is first created, before it appears: use it for set-up like Randomize, clearing labels and disabling buttons. **FormActivate** happens **every time** the form becomes the active window (for example when you return to it): use it to refresh something or to put the cursor in an edit box.

Delphi

```
procedure TForm1.FormCreate(Sender: TObject);
begin
  Randomize;                    // once only
  lblResult.Caption := '';
  btnCalc.Enabled := False;
end;

procedure TForm1.FormActivate(Sender: TObject);
begin
  edtName.SetFocus;             // every time the form is activated
end;
```

5.7 FormCreate or FormActivate? [4]

Which event is the better place for each job?

1	Call Randomize once when the program starts.	**FormCreate**	**FormActivate**
2	Put the cursor in edtName each time the user returns to the form.	**FormCreate**	**FormActivate**
3	Set lblResult to an empty string when the program opens.	**FormCreate**	**FormActivate**
4	Refresh the “last visited” label each time the form comes to the front.	**FormCreate**	**FormActivate**

UNIT 6 Devices, ports and networks

No program runs alone. It runs on hardware, and today most computers are linked together. This theory unit uses diagrams and classification, with no computer needed.

By the end of this unit I can: (tick each box when you are sure)

- ☐ classify devices as input, output or storage
- ☐ name the main parts inside the system unit and say what each does
- ☐ match common ports to their jobs
- ☐ give reasons why computers are networked
- ☐ classify networks as PAN, HAN, LAN or WAN
- ☐ compare client-server and peer-to-peer networks and draw a school network

One thing I want to ask my teacher: _____

THINK FIRST · no computer needed

6.1 One printer, forty computers

A school has 40 computers in two labs, one good printer, and one internet line. Nothing is connected. List **three problems** this causes. Then list **three things** that become possible if all the computers are joined together.

6.2 Input, output or storage? [8]

Circle the category for each device.

1	Scanner	**Input**	**Output**	**Storage**
2	Data projector	**Input**	**Output**	**Storage**
3	Solid state drive (SSD)	**Input**	**Output**	**Storage**
4	Microphone	**Input**	**Output**	**Storage**
5	Speakers	**Input**	**Output**	**Storage**
6	Memory card	**Input**	**Output**	**Storage**
7	Barcode scanner at the till	**Input**	**Output**	**Storage**
8	Printer	**Input**	**Output**	**Storage**

LEARN IT

Inside the system unit

Think of the system unit as a **school**. The **motherboard** is the school building and its corridors: the main circuit board that every part plugs into. The **CPU** is the teacher: it carries out the instructions of every program (speed in GHz). **RAM** is the desks: temporary working space for the programs and files in use right now, and it is **emptied when the power goes off**. **Storage** (hard disk or SSD) is the library: the files are kept there permanently, even when the computer is off. **Ports** are the doors where cables and devices connect.

6.3 Match the part to its job [4]

Write the name of the part: motherboard, CPU, RAM or storage.

Job	Which part? (write the name)
Carries out program instructions; the "brain"	
The main circuit board that joins all the parts together	
Temporary workspace that is wiped when the power is off	
Keeps your files when the computer is switched off	

6.4 Ports [4]

Write what each port is used for.

Port	Used for
USB	
HDMI	
Network (NIC) port	
VGA	

LEARN IT

Why network, and how big is the network?

A **network** is two or more devices linked so that they can share. People network computers to **share** printers, files and one internet connection, to **communicate** (email, messaging) and to keep data and **backups** in one place. Networks are named by size: **PAN** (Personal Area Network): around one person, a few metres, such as Bluetooth earbuds. **HAN** (Home Area Network): inside one home, such as the Wi-Fi router joining phones, TV and laptop. **LAN** (Local Area Network): one building or campus, such as a school computer lab. **WAN** (Wide Area Network): towns, countries, continents, such as the Internet or a bank's network of branches and ATMs.

6.5 PAN, HAN, LAN or WAN? [6]

Circle the best answer.

1	A smartwatch linked to your phone by Bluetooth	**PAN** **HAN** **LAN** **WAN**
2	The Wi-Fi at your house connecting the family's phones and the smart TV	**PAN** **HAN** **LAN** **WAN**
3	The computer lab at your school	**PAN** **HAN** **LAN** **WAN**
4	A bank's ATMs and branches across South Africa	**PAN** **HAN** **LAN** **WAN**
5	Two buildings on the same school campus joined by a cable	**PAN** **HAN** **LAN** **WAN**
6	The Internet	**PAN** **HAN** **LAN** **WAN**

LEARN IT

Client-server and peer-to-peer

In a **client-server** network one powerful computer, the **server**, controls the network: it holds the files, manages log-ins and security, runs the printer and does the backups. The other computers are **clients**. It is expensive and the administrator has control, but if the server fails everybody is affected. In a **peer-to-peer (P2P)** network every computer has **equal** status: any can share files or a printer, and there is no central computer. It is cheap and simple, but security and backups are handled on each computer, so it suits only a few devices.

6.6 Client-server or peer-to-peer? [5]

Circle the type that fits each description.

1	A school computer lab where one server controls all log-ins	**Client-server** **Peer-to-peer**
2	Two friends share files between their laptops at home	**Client-server** **Peer-to-peer**
3	One administrator manages every user account and the backups	**Client-server** **Peer-to-peer**
4	Every computer has equal status and there is no central computer	**Client-server** **Peer-to-peer**
5	If one computer fails, everybody loses access to the shared files	**Client-server** **Peer-to-peer**

6.7 Draw a school network [8]

Draw a **star** network for a small computer lab on the blank page below. Show: **5 PCs** (one is the teacher's), a **server**, a **printer**, a **switch** in the middle, and a **router** linking the switch to the **Internet**. Label every device. Under the drawing write what type of network the lab is (PAN, HAN, LAN or WAN) and whether it is client-server or peer-to-peer.

6.8 Recommend a network [3]

A spaza shop owner has three computers (the till, an office laptop and a tablet) and one printer. She asks you whether she needs a server. Recommend **client-server** or **peer-to-peer** and give **two** reasons.

TERM CHECKPOINT

Decisions, text and networks

Time: 60 minutes	**Total:** 55 marks	**Mark:** _____ / 55

No notes, no computer. Show all your working. Write Delphi code neatly and use correct indentation.

SECTION A — Trace and predict [12]

A1 [6]

A school fun run charges an entry fee. Trace the code for each runner. Write the value of **iFee** after the age tests, and the final iFee after the member discount.

```delphi
if iAge < 6 then
  iFee := 0
else
  if iAge <= 17 then
    iFee := 30
  else
    iFee := 60;
if bMember and (iFee > 0) then
  iFee := iFee - 10;
```

iAge	bMember	iFee after age tests	Final iFee
4	True		
12	False		
17	True		

A2 [3]

What appears in **lblOut**?

```delphi
sTxt := '  Load shedding  ';
sTxt := Trim(sTxt);
iLen := Length(sTxt);
sTxt := UpperCase(sTxt);
lblOut.Caption := IntToStr(iLen) + ' ' + sTxt;
```

A3 [3]

Write the value of **sSize** after the code runs for **iPies = 0**, for **iPies = 4** and for **iPies = 9**.

Delphi

```
case iPies of
  0    : sSize := 'None';
  1, 2 : sSize := 'Small';
  3..5 : sSize := 'Medium';
else
  sSize := 'Big';
end;
```

SECTION B — Concepts [15]

B1 [4]

Complete the truth table.

A	B	(NOT A) OR B
T	T	
T	F	
F	T	
F	F	

B2 [3]

Circle PAN, HAN, LAN or WAN for each.

1	Wireless earphones paired with a phone	**PAN**	**HAN**	**LAN**	**WAN**
2	The network that links all computers in the school lab	**PAN**	**HAN**	**LAN**	**WAN**
3	The network that links a cellphone company's towers across the country	**PAN**	**HAN**	**LAN**	**WAN**

B3 [3]

Circle Input, Output or Storage for each device.

1	Webcam	**Input**	**Output**	**Storage**
2	Flash drive	**Input**	**Output**	**Storage**
3	Monitor	**Input**	**Output**	**Storage**

B4 [3]

The _____ is the main circuit board that joins all the parts. The _____ carries out instructions. _____ is the temporary workspace that is wiped when the power goes off.

B5 [2]

Circle the network type described.

1	All the computers have equal status and there is no central server.	**Client-server**	**Peer-to-peer**
2	A central computer controls log-ins and the printer.	**Client-server**	**Peer-to-peer**

SECTION C — Find the errors [8]

C1 [4]

Find **two** errors and write the corrected code.

Delphi

```
if iMark >= 0 and iMark <= 100 then
  lblOk.Caption := 'Valid';
else
  lblOk.Caption := 'Invalid';
```

C2 [4]

Find **two** errors in this CASE statement.

Delphi

```
case iMark of
  0..50   : sResult := 'Fail';
  50..100 : sResult := 'Pass';
else
  sResult := 'Invalid';
```

SECTION D — Write the code [20]

D1 [8]

A quiz mark out of 20 is in the spin edit **sedScore**. If the score is below 0 or above 20 show *Invalid score* with ShowMessage and stop. Otherwise use a CASE statement to put a medal in **lblMedal**: 18 to 20 *Gold*, 15 to 17 *Silver*, 10 to 14 *Bronze*, anything lower *Keep trying*.

D2 [5]

The learner types a name in **edtName**. Trim it. If it is empty show *Please enter a name* with ShowMessage. Otherwise display *Hello,* followed by the name in capitals in **lblWelcome** and the number of characters in the name in **lblLen**.

D3 **[4]**

When **btnClear** is clicked, ask *Delete this record?* with a confirmation icon and Yes/No buttons. If the user clicks Yes clear **memOut** and show *Cleared* with ShowMessage. (Write the whole event handler.)

D4 **[3]**

Write one IF statement (the whole statement) that shows *Free entry* in **lblEntry** when **iAge** is under 6 **or** is 60 and over, and *Pay at the gate* otherwise.

COMING UP NEXT TERM · warm-up

Thirty scripts to mark

In Term 3 you will make the computer do something many times without you writing it many times. Before the Delphi, think about a teacher with a pile of **30 test scripts** to mark.

(a) Write, in order, the steps Mr Naidoo does for **one** script. (b) Which steps are done again for every script, and which are done only once (before he starts or after he finishes)? (c) What tells him when to **stop**? (d) What if nobody told him how many scripts there were? How would he know the pile is finished?